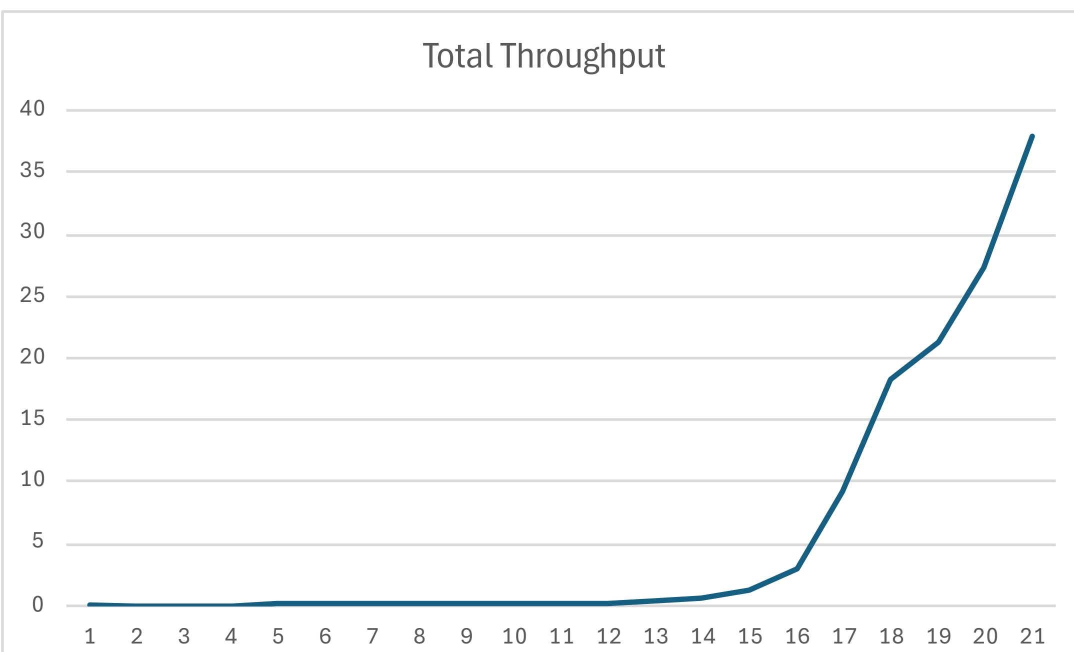
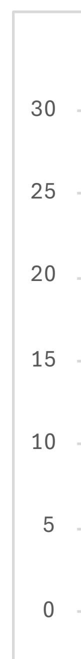
Average Turnaround	Total Throughput	Average CPU Utilization	Processes in Ready Queue
10	0.0685342	10.2123	69.9891
11	0.0695913	10.9422	76.1483
12	0.0750342	11.7934	88.4907
13	0.0808031	12.8988	104.226
14	0.0869129	14.1213	122.733
15	0.0970324	15.0508	146.042
16	0.111035	16.0689	178.422
17	0.140653	17.1339	240.993
18	0.14511	18.0089	261.328
19	0.17753	19.1759	340.43
20	0.183089	19.9668	365.568
21	0.268056	21.1091	565.842
22	0.32972	21.8538	720.564
23	0.606431	22.9893	1394.14
24	1.264	23.7901	3007.08
25	2.99953	24.8575	7456.09
26	9.28116	24.5556	22790.4
27	18.3301	24.9095	45659.3
28	21.3824	24.6052	52611.8
29	27.3064	24.9568	68148
30	37.9108	24.8624	94255.2





Total Throughput shows a linear rate that begins to drasticall

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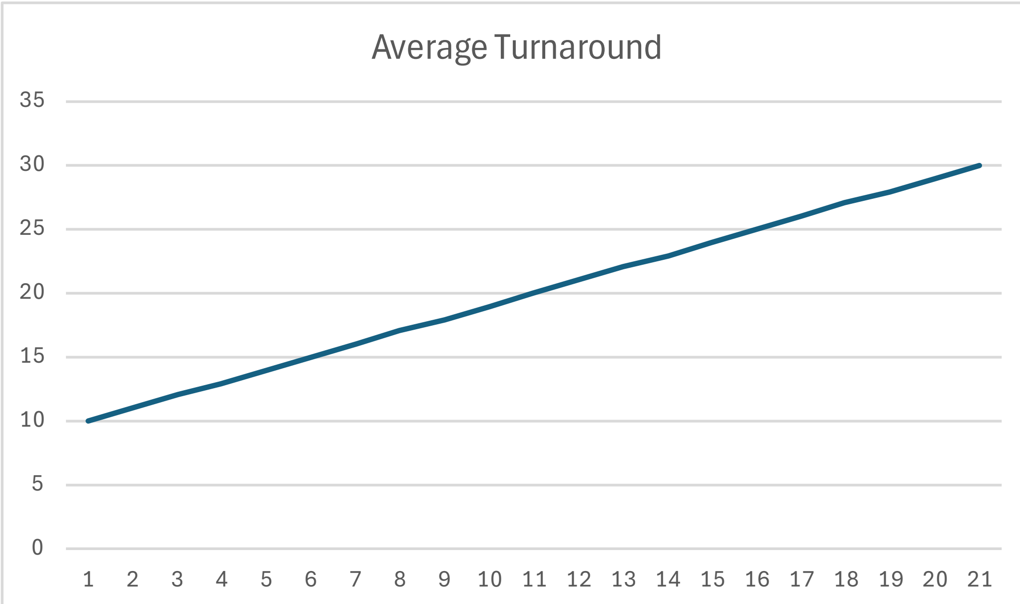
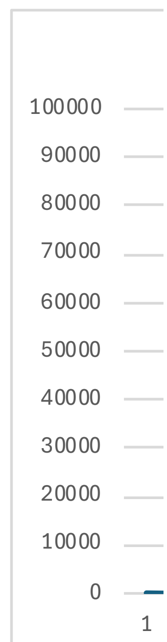
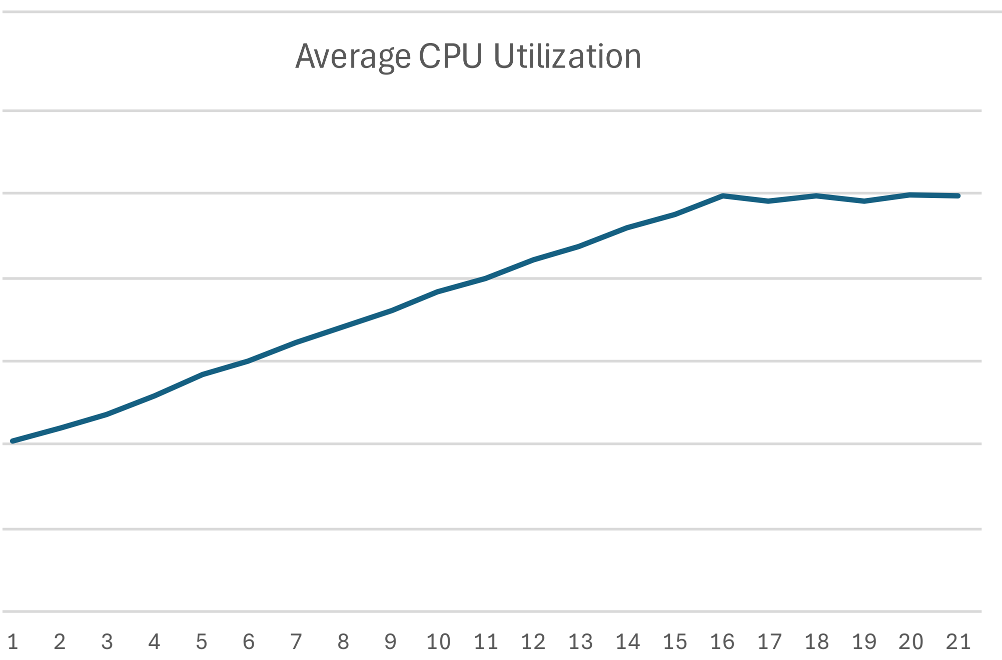


Average Turnaround shows a positive realtionship because the linear line increases with the arrival time





y increase

Average CPU Utilization shows a positive realtionship because the linear line increases

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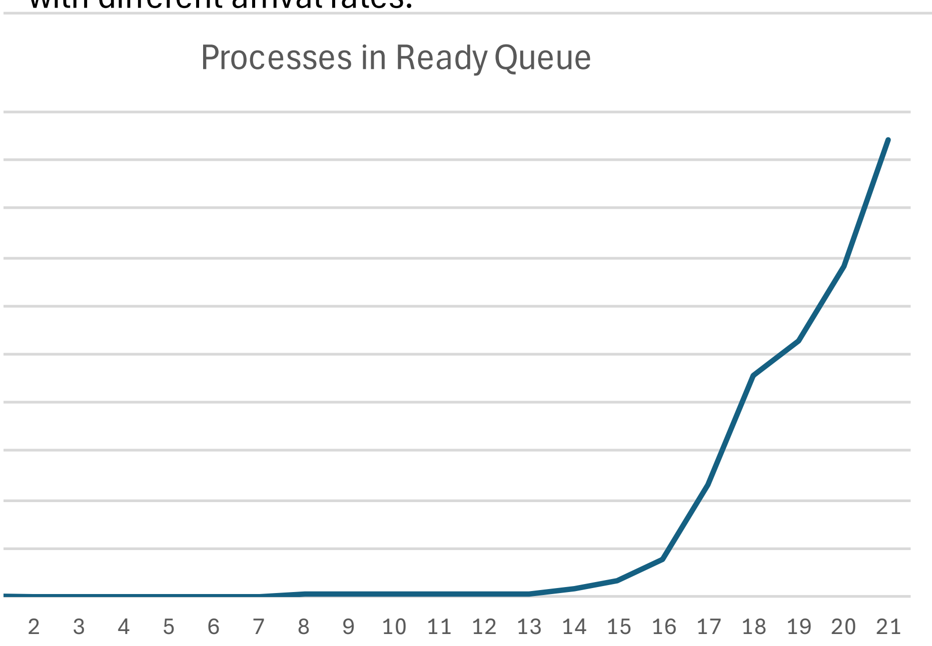
with the arrival time but then it goes into a
unstaeady balance

FL-cs3360-PA3: This file contains the C++ code for the First-Come First-Served CPU scheduling simulation, which generates metrics based on varying arrival rates and service times.

simulation_results.csv: This file stores the output metrics from the simulation runs, including average turnaround time, throughput, CPU utilization, and processes in the ready queue

The program takes two command-line arguments: the average arrival rate (λ) and the average service time. It runs simulations for λ values ranging from 10 to 30 processes per second, generating service times based on an exponential distribution.

The simulation results indicate how average turnaround time, throughput, CPU utilization, and processes in the ready queue vary with different arrival rates.



Processes in Ready Queue shows asteady but positive ealtionship